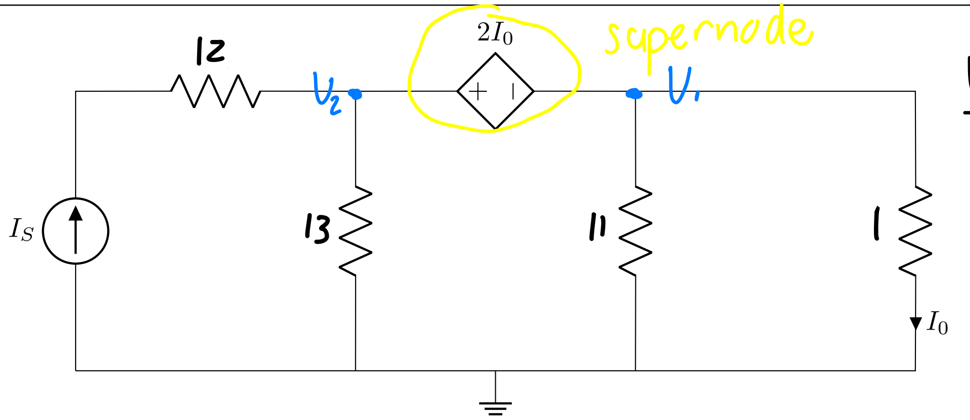
Section 1

Question 1 [2]	
Given: R_1, R_2, R_3, R_4.	



$\frac{V_1}{1} = 1$

$V_1 = 1V$ ①

01	Use the principle of linearity to determine I_0/I_S. [2]

Assume $I_0 = 1A$

Supernode: $V_2 - V_1 = 2I_0$

$V_2 - V_1 = 2$

①: $V_2 - 1 = 2$

$V_2 = 3V$

Nodal: $-I_S + \frac{V_2}{13} + \frac{V_1}{11} + \frac{V_1}{1} = 0$

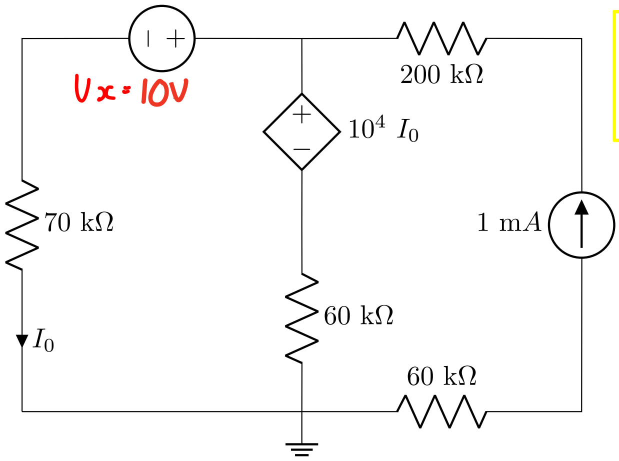
$I_S = \frac{189}{143} A$

$I_0/I_S = \frac{1}{\frac{189}{143}}$

$= \frac{143}{189}$

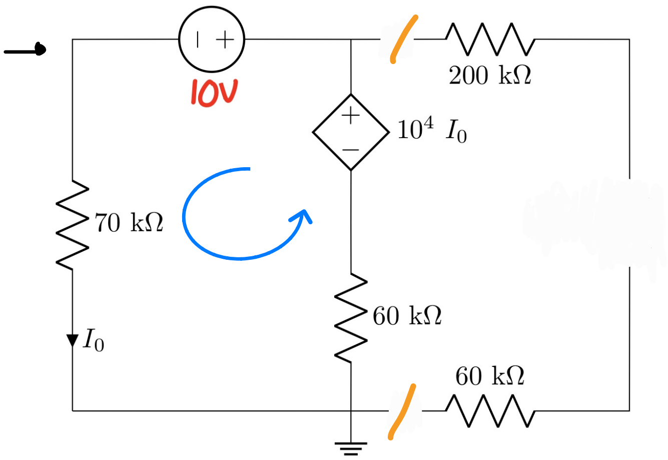
$= 0.75661$

Question 2 [2]	
Given: V_x.	



Voltage: close
Current: remove

02	Find $I_0(V_x)$, the independent contribution of Vx to I_o [2]



KVL: $-10^4 I_0 + 10 + 70\times10^3 I_0 + 60\times10^3 I_0 = 0$

$120 \times 10^3 I_0 = -10$

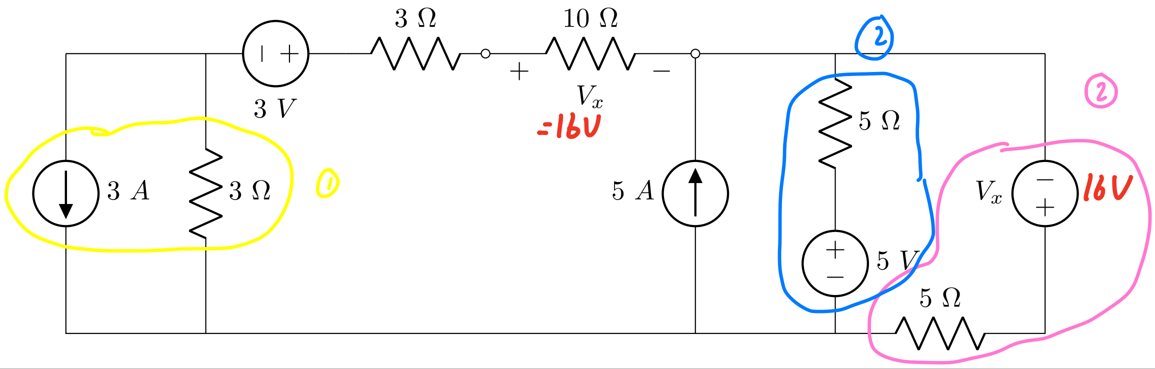
$I_0(V_x) = -83.33\ \mu A$

Question 3 [2]

Given: V_x, V_1, R_1, R_2.

Use source transformation to obtain Circuit B so that it is equivalent to Circuit A. Determine V_2 in Circuit B.

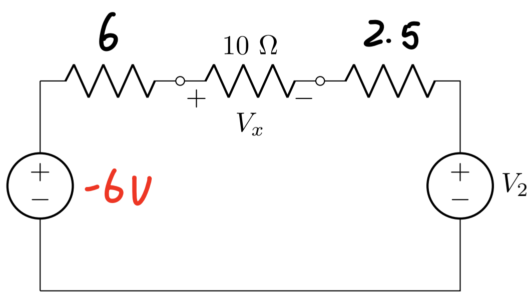
Circuit A:



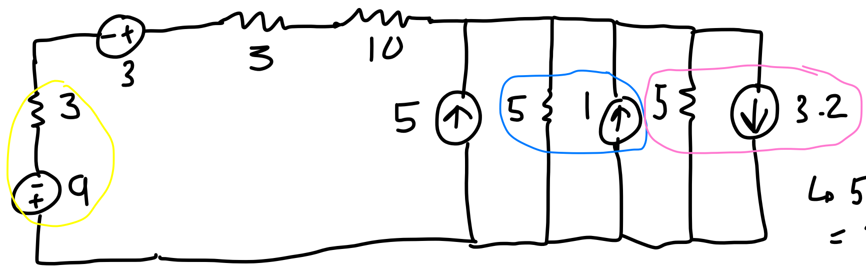
Circuit B:

$V = IR$

$I = \frac{V}{R}$



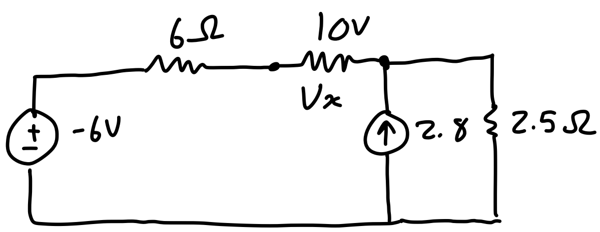
03	V_2 [2]



↳ 5+1 -3.2

= 2.8 A

↳



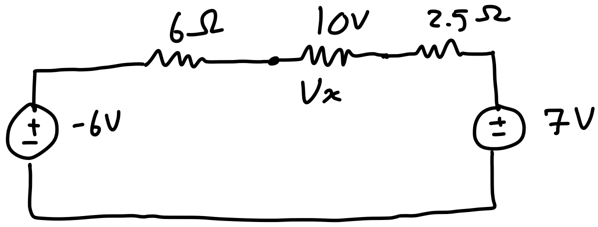
$\frac{5 \times 5}{5} = 2.5\,\Omega$ (Parallel)

2.5×2.8

$= 7$

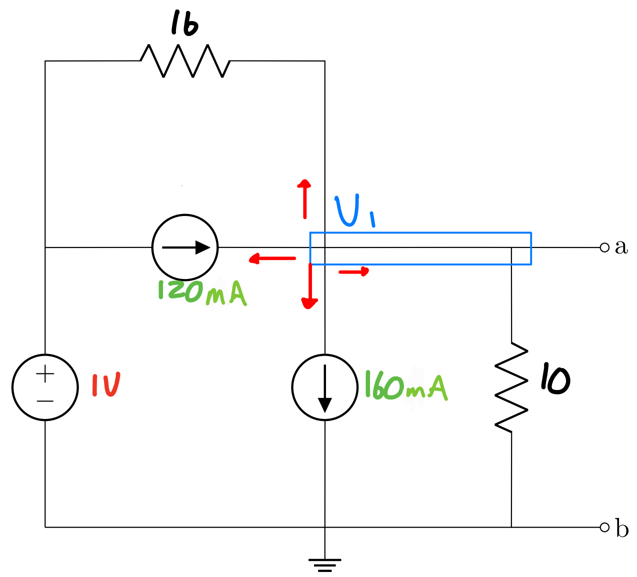
$\therefore V_2 = 7\,V$

↳



Question 4 [2]

Given: $R_1,\ R_2,\ v_1,\ i_1,\ i_2$.



04	Find the Thevenin equivalent voltage V_{Th} with regards to terminals a-b. [2]

Nodal: $\frac{V_1 - 1}{16} + \frac{V_1}{10} - 120\times10^{-3} + 160\times10^{-3} = 0$

$V_1\left(\frac{1}{16} + \frac{1}{10}\right) = -40\times10^{-3} + \frac{1}{16}$

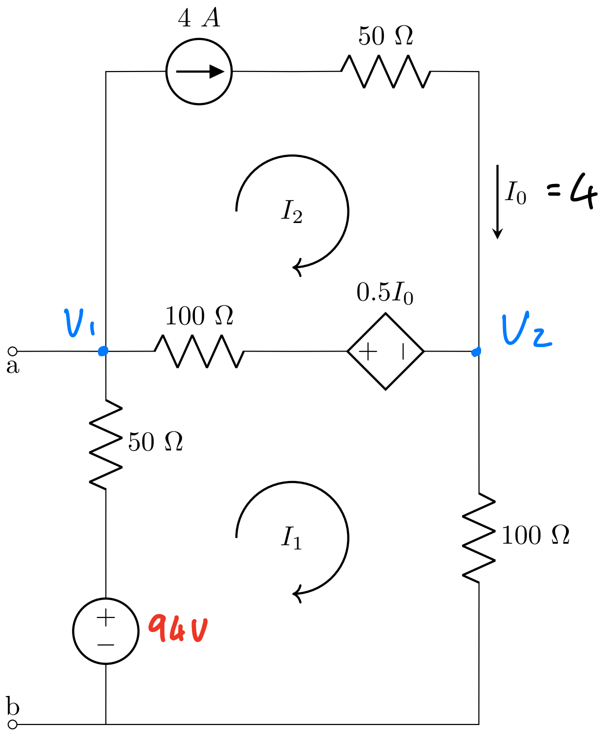
$\frac{13}{80} V_1 = 22.5\times10^{-3}$

$V_1 = V_{ab} = V_{th}$

$V_{th} = 138.462\ mV$

Question 5 [2]

Given: V_x.



05	Find the Thevenin equivalent voltage V_{Th} between terminals a-b. [2]

Nodal @ V_1: $\dfrac{V_1 - 0.5(4) - V_2}{100} + \dfrac{V_1 - 94}{50} + 4 = 0$

$\frac{3}{100} V_1 - \frac{1}{100} V_2 = -2.1$ ①

@ V_2: $\dfrac{V_2 + 0.5(2) - V_1}{100} + \dfrac{V_2}{100} - 4 = 0$

$-\frac{1}{100} V_1 + \frac{1}{50} V_2 = 3.98$ ②

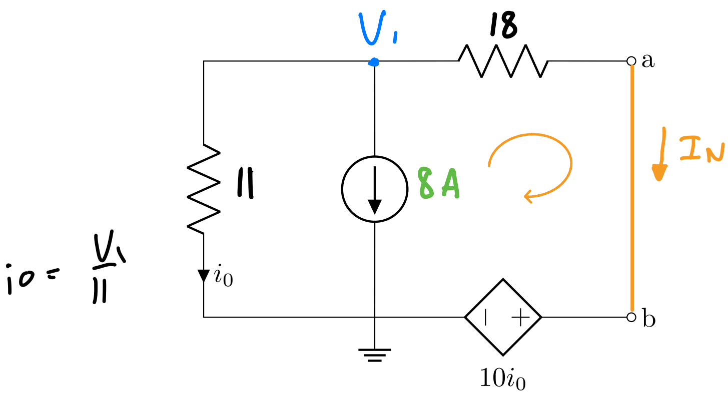
① & ② in Calculator

↳ $V_1 = -4.4$ V, $V_2 = 196.8$ V

$V = IR$

$V_{th} = V_{ab} = V_1$

$= -4.4V$

Questions 6 to 7 [4]	
Given: R_1, R_2, i_1.	
	
06	Determine the Norton equivalent current I_N. [2] if 1A, $I_N = -227.27mA$
07	Determine the Norton equivalent resistance R_N. [2]

⑥ @V_1: $\frac{V_1}{11} + 8 + \frac{V_1 - 10i_0}{18}$ I_n $= 0$

$$\frac{29}{198} V_1 - \frac{5}{9} i_0 = -8$$

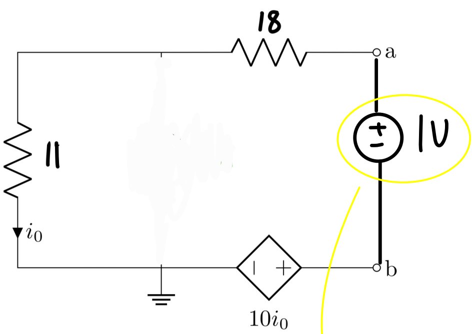
$$\frac{29}{198} V_1 - \frac{5}{99} V_1 = -8$$

$$V_1 = -83.37V$$

$$I_n = \frac{-83.37 - (-\frac{10}{11} 83.37)}{18}$$

$$= -421.06mA$$

⑦

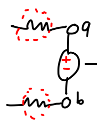


$$29I_0 - 10I_0 - 1 = 0$$

$$I_0 = \frac{1}{19}$$

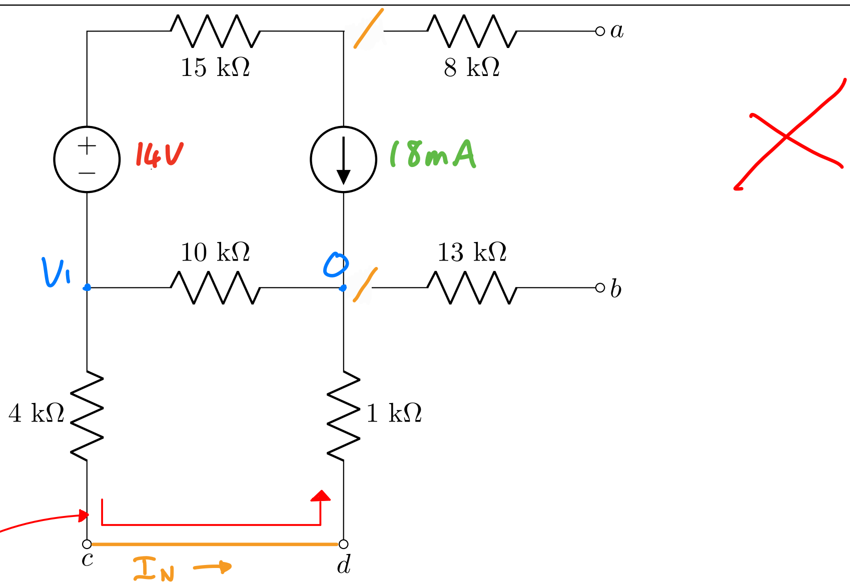
$$R_N = \frac{1}{\frac{1}{19}}$$

$$= 19\Omega$$



→ 1V. If no resistor, use 1A

Question 8 [2]	
Given: V_1, I_1.	



08	Determine the Norton equivalent current I_N between terminals c-d. [2]

@ V_1: $\frac{V_1}{5000} + \frac{V_1}{1000} + 18\times10^{-3} = 0$

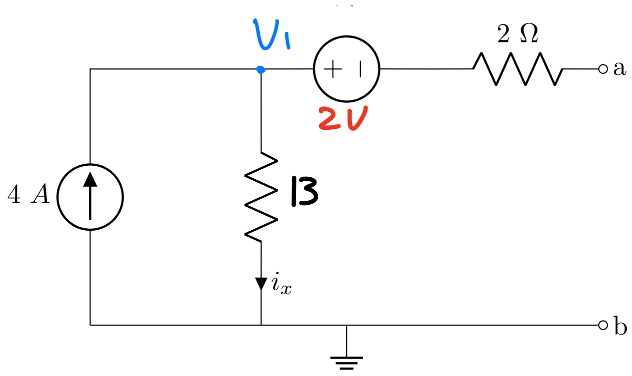
$V_1 = -108V$

$I_N = \frac{V_1}{5000}$

$= \frac{-108}{5000}$

$= -21.6\times10^{-3}\ mA$

Questions 9 to 11 [4]	
Given: R_1, V_x.	
09	Find the Thevenin equivalent voltage V_{Th} across terminals a-b. [2]
10	Find the Thevenin equivalent resistance R_{Th} across terminals a-b. [1]
11	Calculate the maximum power that can be transferred to a load resistor R_L that can be connected between terminals a-b. [1]



(9) @ V_1: $\frac{V_1}{13} - 4 = 0$

$V_1 = 52\,V$

$V_1 - V_{ab} = 2$

$52 - V_{ab} = 2$

$V_{ab} = V_{th} = 50\,V$

(10) 

$= 15\,\Omega$

(11) $P_{max} = \frac{V_{th}^2}{4R_{th}}$

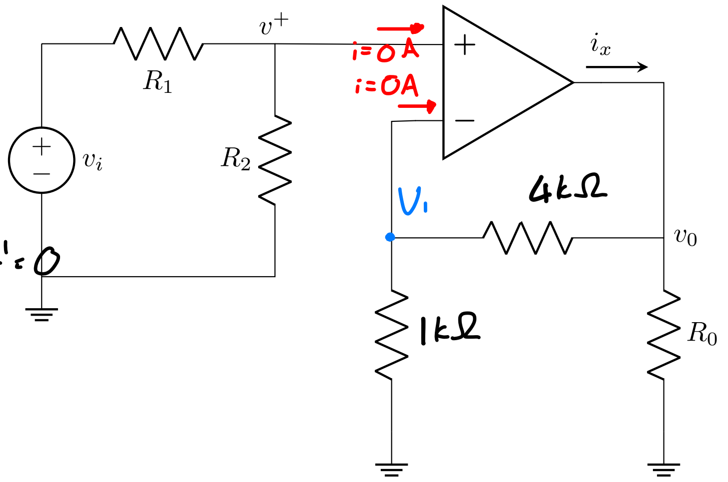
$= \frac{50^2}{60}$

$= 41.667\,W$

Questions 12 to 13 [2]

Given: R_3, R_4.

$$\frac{V_1 - V_o}{4} + \frac{V_1}{1} = 0$$



$V_1 = v^+$

Given: R_0, v^+ $R_0 = 41\Omega$, $v^+ = 3V$	
12	Find the current i_x. [1] ✗
Given: $R_1 = R_2$	
13	Calculate the closed-loop gain (v_0/v_i) of the circuit. [1]

(12) $V_1 = v^+ = 3V$

@ V_1: $\frac{V_1}{1000} + \frac{V_1 - V_o}{4000} = 0$

$5V_1 = V_o$

$V_o = 15V$

@ V_o: $\frac{V_o - V_1}{4000} + \frac{V_o}{41} - i_x = 0$

$i_x = \frac{15-3}{4000} + \frac{15}{41}$

$= 368.85mA$

(13) @ v^+: $\frac{V^+ - V_i}{R_1} + \frac{V^+}{R_2} = 0$

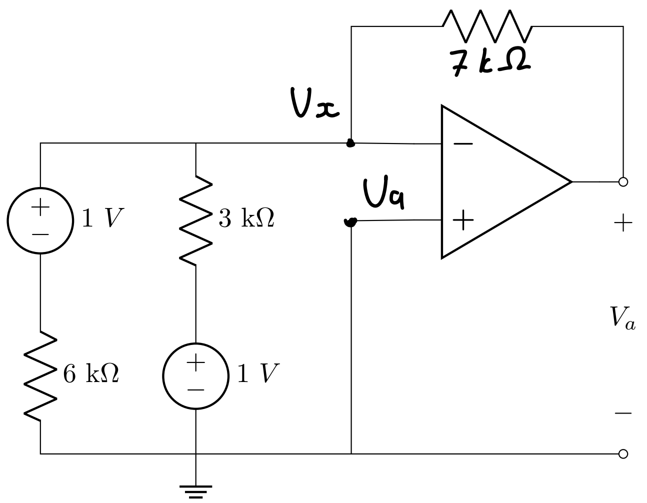
$2V^+ = V_i$

$V_i = 6V$

$\frac{V_o}{V_i} = \frac{15}{6}$

$= 2.5$

Question 14 [1]	
Given: R_z.	



$(V_x = V_a)$

14	Determine V_a. [1]

@ V_x: $\frac{V_x - 1}{6\times10^3} + \frac{V_x - 1}{3\times10^3} + \frac{V_x - V_a}{7\times10^3} = 0$

$V_x = V_a$

$V_a\left(\frac{1}{6k} + \frac{1}{3k}\right) = \frac{1}{6k} + \frac{1}{3k}$

$\underline{V_a = 1\ V}$

$7V_x - 7 + 14V_x - 14 + 6V_x - 6V_a = 0$

$27V_x - 6V_a = 21$

$V_a = 1\ V$

Total Section 1: [23]

Grand Total : [23]